

# Lecture 01

## Introducing Today Technology

Information Technology Fundamentals  
IT-06016402

Prof. Dr. Kitsuchart Pasupa



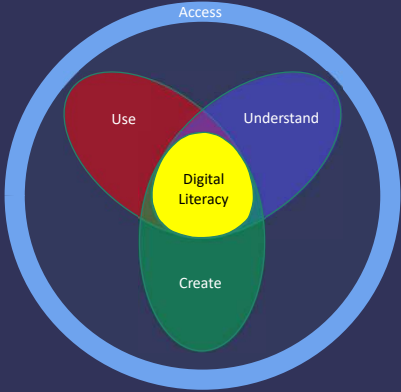
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# Today Technologies

- Technology changes everyday, you must keep up with these changes to remain digitally literate
- Digital literacy involves having a current knowledge and understanding of computers, mobile devices, the web, and related technologies



<https://www.eetimes.com/sensors-technology-of-today-and-tomorrow>



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# Computer

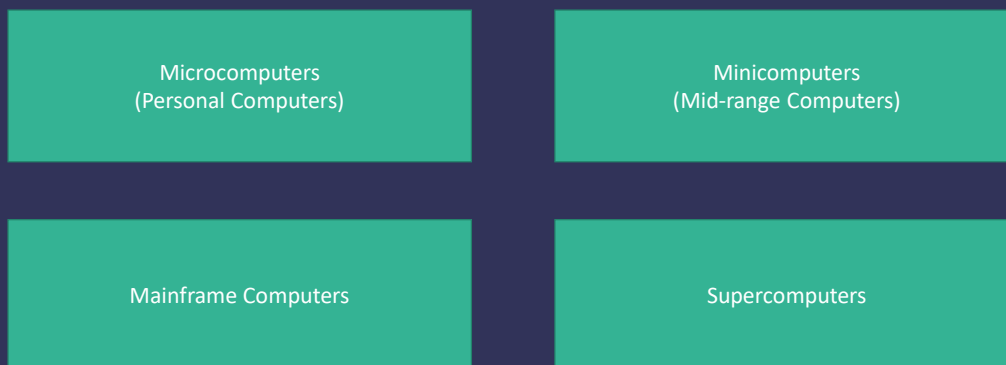
- Computer contains many electric and electronic components (and might contains mechanical components) known as hardware
- It operates under the control of instructions stored in its own memory, known as software.



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# Class of Computer

- It can be classified in many ways, such as by purpose, function, usage, generation of technology
- Classified by purpose on the basis of size



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# Data & Information

**Data**

- It is a raw and unorganized fact that required to be processed to make it meaningful.
- It comprises facts, observations, perceptions numbers, characters, symbols, image, etc.

**Information**

- Information is a set of data which is processed and structured in a meaningful way according to the given requirement.

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# Data & Information

|                     |             |     |
|---------------------|-------------|-----|
| Medium Sodas        | ฿60.0 each  | x 2 |
| Small Chicken Wings | ฿100.0 each | x 2 |
| Caesar Salad        | ฿150.0 each | x 1 |
| Cookies             | ฿75 each    | x 3 |
| Amount Received     | ฿1,000      |     |

Data (Input)

Process/Store

|                                                |                     |         |
|------------------------------------------------|---------------------|---------|
| IT.KMITL SuperStore                            |                     |         |
| 1 Soi Chalongsong 1, Ladkrabang, Bangkok 10520 |                     |         |
| Tel: +66 2 723 4900                            |                     |         |
| Quantity                                       | Item                | Total   |
| 2                                              | Medium Sodas        | 120.00  |
| 2                                              | Small Chicken Wings | 200.00  |
| 1                                              | Caesar Salad        | 150.00  |
| 3                                              | Cookies             | 225.00  |
| Total Due                                      |                     | 695.00  |
| Amount Received                                |                     | 1000.00 |
| Change                                         |                     | 305.00  |
| Thank You ☺                                    |                     |         |

Information (Output/Receipt)



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# Data & Information—Input

- A keyboard contains keys you press to enter data and instructions into a computer or mobile device



Desktop Keyboard



Laptop Keyboard



Onscreen Keyboard



Type Cover Keyboard



Mini Keyboard



Virtual Keyboard

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# Data & Information—Input

- A pointing device is an input device that allows a user to control a small symbol on the screen called the pointer.



=





Trackball - Tom Cranston, Fred Longstaff and Kenyon Taylor (1952)



Mouseketeer - Douglas Engelbart (1963)



Xerox Alto Mouse (1973)

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# Data & Information—Input

Some devices enable you to speak data instructions using **voice** input



Microphone

Built-in Microphone

Ear Phone

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# Data & Information—Input

Some devices enable you to capture live full-motion images using **video** input



Webcam

Built-in Webcam

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# Data & Information—Input

Some devices enable you to capture images



Scanner



Digital Camera

A scanner is a light-sensing input device that converts printed text and images into a form the computer can process

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# Data & Information—Output

- An output device is any hardware component that conveys information from a computer or mobile device to one or more people

Printer

An output device that produces text and graphics on a physical medium, such as paper or other material



3D Printer

A 3-D printer can print solid objects, such as implants, toys, parts, prototypes, and more



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# Data & Information—Output

- A display is an output device that visually conveys text, graphics, and video information



Desktop Display



Laptop Display



Tablet Display



Smartphone Display



Digital Camera Display

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# Data & Information—Output

Speakers allow user to hear audio such as music, voice, and other sounds



Headphone



Earbud



Ear Phone

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# Data & Information—Storage

- Memory consists of electronic components that store instructions waiting to be executed and the data needed by those instructions
- Storage media is used to keep data, instructions, and information. These are operated by a storage device (writes/reads).



Harddisk Drive (HDD)



USB Flash Drive



Optical Disc



Solid State Drive (SSD)



Memory Card



Cloud Storage

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# The Web

- The Internet is a worldwide collection of computer networks that connects millions of businesses, government agencies, educational institutions, and individuals
- The Internet was first invented for military purposes, and then expanded to the purpose of communication among scientists under the development of the Advanced Research Projects Agency Network (ARPANET) in the late 1960s.
  - University of California, Los Angeles (UCLA)
  - Stanford Research Institute's Augmentation Research Center
  - University of California, Santa Barbara (UCSB)
  - University of Utah

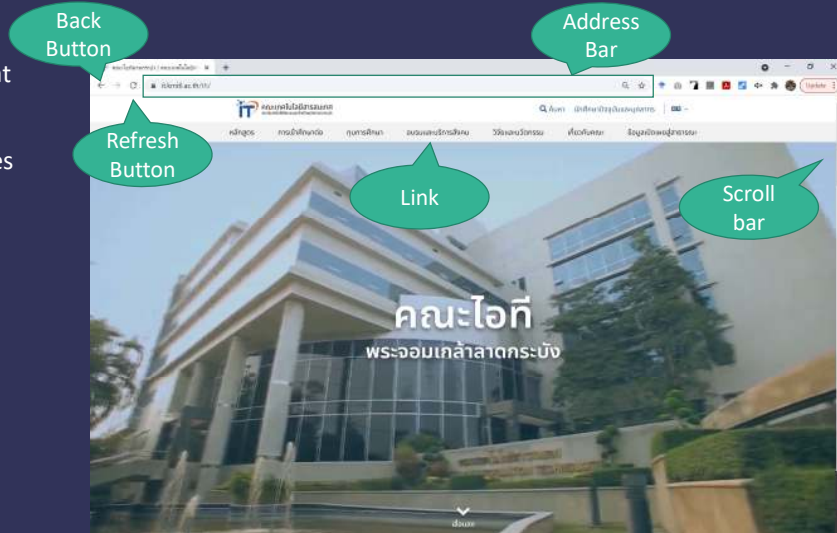


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# Surf the Internet—Web Browser

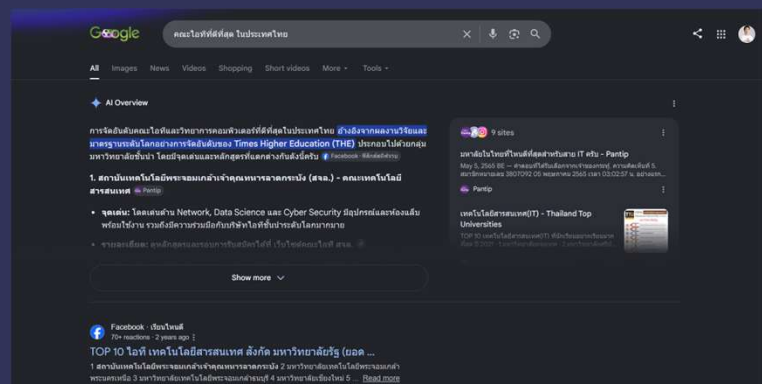
- A browser is software that enables users with an Internet connection to access and view webpages on a computer or mobile device.



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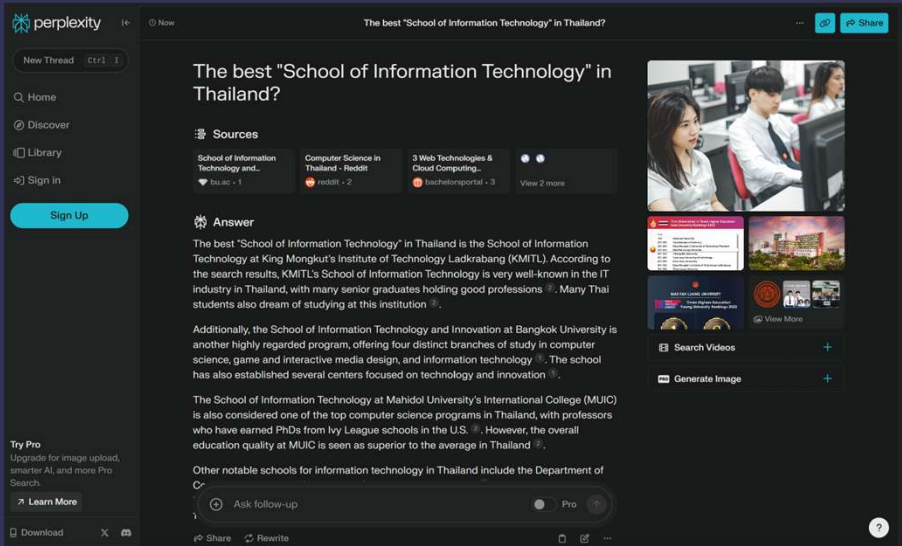
# Looking for something on the Internet?

- A search engine is software that finds websites, webpages, images, videos, news, maps, and other information related to a specific topic
- It collects the website by using spiderbot or crawler. The crawler is an Internet bot that systematically browses the World Wide Web for the purpose of Web indexing.



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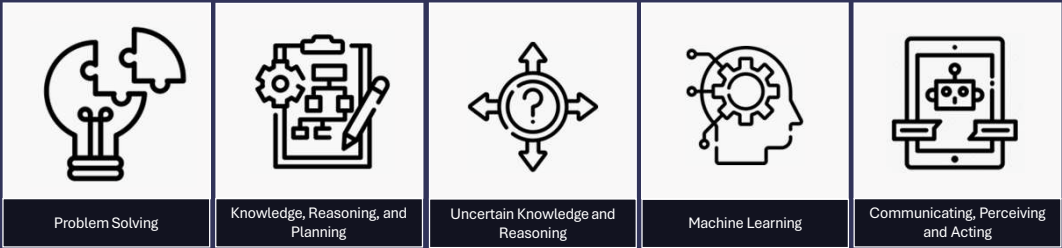
# AI Search Tool



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# Artificial Intelligence

- Thinking and acting humanly and rationally
- The use of computer science, engineer, cognitive sciences and others, in conjunction with computer programming
- Computer can possess like human, e.g., image recognition, speech recognition



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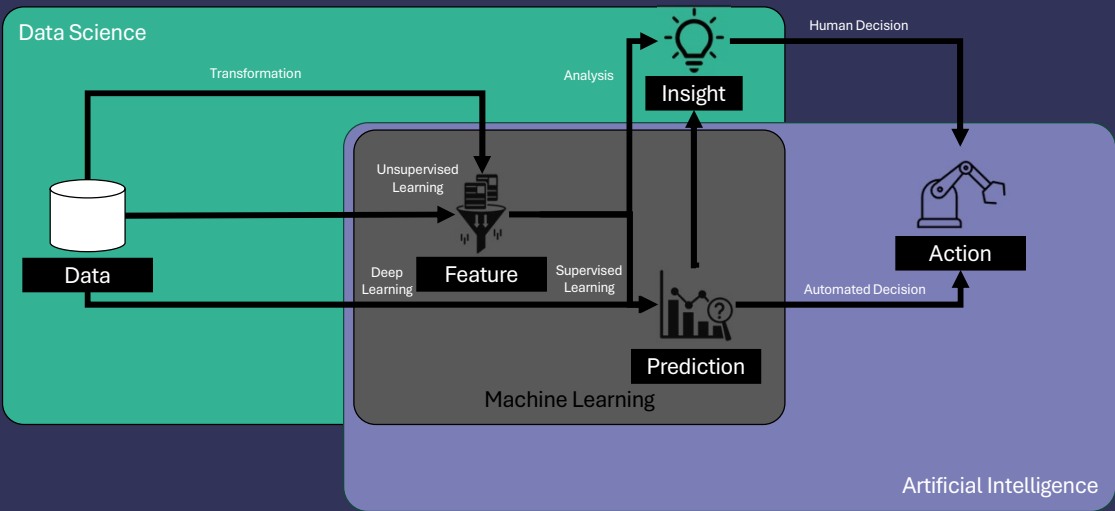
# Data

- Types of data
  - Structured – tables, numerical data, logs, etc.
  - Unstructured – images, audio, text, etc.
- Data is imperfect – may include incorrect labels or missing values.
- Simply collecting data is not enough – refining and validating it is essential before it becomes truly valuable for AI development.



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# AI, Data Science, ML Relationship



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# Type of AI – Based on Capabilities

Artificial Narrow Intelligence - Narrow range of abilities

- The only AI that exists today
- Trained for a single or narrow task, often faster and better than humans
- Cannot perform tasks outside its defined scope
- Examples: Siri, Alexa, IBM Watson, OpenAI ChatGPT

Artificial General Intelligence - On par with human capabilities

- Currently theoretical
- Can learn and perform any intellectual task that a human can
- Uses prior knowledge to solve new problems without retraining
- Represents true human-level machine intelligence

Artificial Super Intelligence - More capable than a human

- Strictly theoretical
- Would surpass human cognitive abilities
- Could think, reason, learn, make judgments, and experience emotions
- Represents AI evolution beyond human comprehension

Artificial Narrow Intelligence

ANI



Artificial General Intelligence

AGI



Artificial Super Intelligence

ASI



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# AI



Anthropic



Google



OpenAI



Perplexity AI



X.AI



Mistral AI



MiniMax



Z.ai



Alibaba Cloud



Deepseek



Meta AI

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# Online Social Network

- An online social network encourages members to share their interests, ideas, stories, photos, music, and videos with other registered users



The screenshot shows a Facebook profile page for 'Faculty of Information Technology, KMITL'. A green callout bubble labeled 'Search Bar' points to the search bar at the top of the page. Another green callout bubble labeled 'Create a post' points to the 'Create Post' button on the right side of the page.

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# Online Social Network Platforms

|                                                                                     |                                                                                     |                                                                                      |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  |  |
| Facebook                                                                            | Instagram                                                                           | Linkedin                                                                             | Medium                                                                                |
|  |  |  |  |
| X                                                                                   | Clubhouse                                                                           | Youtube                                                                              | TikTok                                                                                |

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# Digital Safety and Security

- It is very important that all users should protect their computers and mobile devices

Virus and Other Malwares

Privacy

Health Concern

Environmental Issues

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# Virus and Other Malwares

### Virus

- a piece of code that inserts itself into an application and executes when the app is run.

### Ransomware

- disables victim's access to data until ransom is paid

### Spyware

- collects user activity data without their knowledge

### Adware

- serves unwanted advertisements

### Trojans

- disguises itself as desirable code

### Worms

- spreads through a network by replicating itself

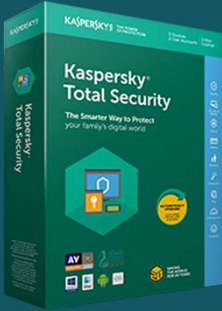
### Keyloggers

- monitors users' keystrokes

# Antivirus









# Digital Safety and Security

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# Privacy

- Crimes Opportunities
  - Social Media is fostering a big rise in real-world stalking
- Cybercrimes Opportunities
  - identity theft
  - financial fraud
  - revealing of personal information—messages, images, and video and audio recordings—without individuals' consent or permission (cyberstalking, cyberharassment/cyberbullying)



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# Privacy—Password

- It keeps your personal information safe (including emails, files, other content) and prevent someone else from getting into your account
- Create a **strong, long, and unique** password that can be memorable to you but nearly impossible for someone else to guess
  - Use a combination of letters, numbers, and symbols (ASCII-standard characters only)
  - Avoid personal information and common words
- Hide written passwords
- Manage your passwords with a tool



# Privacy



# Digital Safety and Security

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# Health Concern

## Technology addition

- Someone becomes obsessed with using technology. Tech dependence can lead to teen consequences that span from mild annoyance when away from technology to feelings of isolation, extreme anxiety, and depression.



## Technology overload

- It is the device proliferation and/or information overload that causes cognitive and physical burdens on human beings due to the use of multiple gadgets with multiple functions to accomplish multiple tasks in everyday activities. You may feel distracted, anxious, fatigued, or even depressed



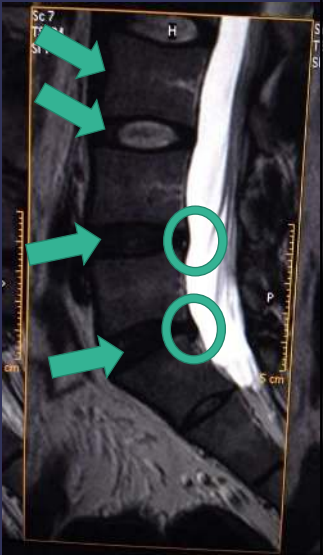
## Office syndrome

- Office Syndrome refers to a group of various symptoms including the inflammation of neck, shoulder and back muscles. These symptoms are caused due to repetitive muscle strain during having long period of unhealthy positions such as sitting in front of a computer screen for several hours.



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# Office Syndrome



Do

Don't

Ergonomic Chair

Spinal Disc Herniation

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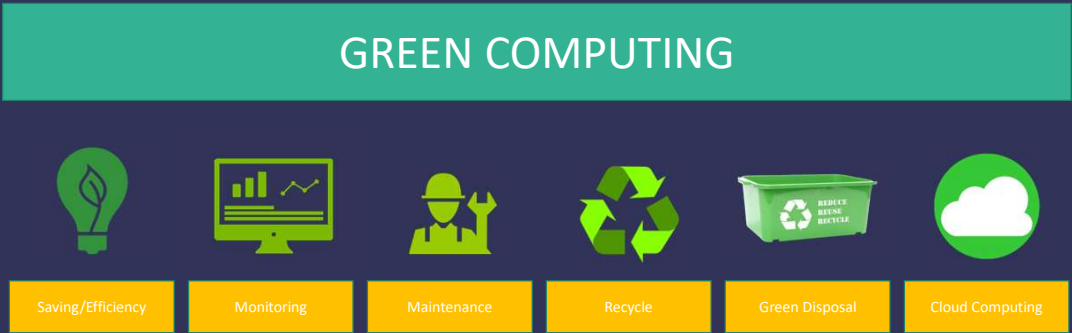
Health Concern

Environmental Issues

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# Environmental Issues

- Green computing is an approach to reduce the electricity consumption and environmental waste generated when using a computer



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# Programs & Apps

- The software known as a program, tells the computer which tasks do it required to perform and how to perform them

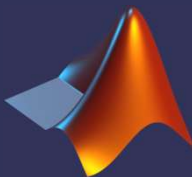


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# Programs & Apps

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Application Software



Office 365    Adobe Creative Cloud    Matlab

User

Applications

Operating Systems

Hardware

↑ ↓   ↑ ↓   ↑ ↓

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# Communications and Networks

- Information can be generated by, one or more of these communications technologies:

Internet    E-mail    Instant Messaging    Chat Room

VoIP    Video Conference    News Group    Fax

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# Communications and Networks

- A communications device is hardware that is capable of transferring items from computers and devices to transmission media and vice versa
- A network is a collection of computers and devices connected together, often wirelessly, via communications devices and transmission media



Cellular Radio



Wi-fi



Bluetooth





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# Technology Uses

Education

Finance

Retail

Government

Entertainment

Healthcare

Research

Manufacturing

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# Technology Users



Home User



Small Home/Office User



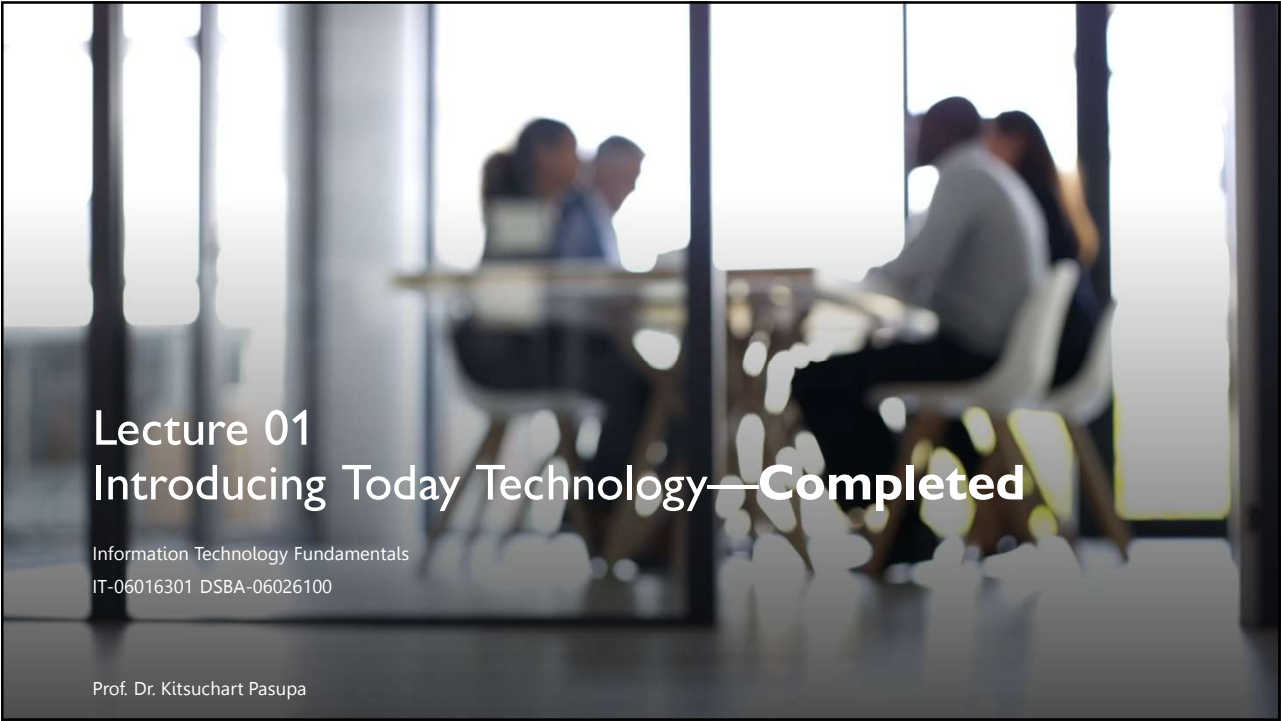
Mobile User



Power User



Enterprise User



## Lecture 01

# Introducing Today Technology—Completed

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IT-06016301 DSBA-06026100

Prof. Dr. Kitsuchart Pasupa